



OPTIMA

AUTOMATED STUDY FLOW

INSTALLATION GUIDE

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Installation Guide

A start-to-finish walkthrough for getting Optima running on a fresh computer. Total time (end-to-end) would be around 5 minutes once Python is installed.

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System requirements

	Minimum	Recommended
Operating system	macOS 12	macOS 14+
Python	3.11	3.13
RAM	2GB free	4GB free
Disk	120 MB	500 MB (leaves sufficient room for personal data)
Display	1280 x 720	1440 x 900 or higher
Internet	Only for first-time install (downloading Flask + pywebview)	—

PRIVACY NOTE

Optima runs entirely on your computer. No data is ever sent to a server. Once installed, it works completely offline.

Step 1 — Installing Python 3.11+

Check whether Python is already installed by opening Terminal and running:

```
python3 --version
```

If you see Python 3.11.x or higher, skip to step 2.

On macOS

1. Open the Terminal application (⌘Space → type "Terminal")
2. If you have Homebrew installed: brew install python@3.13
3. Otherwise, download the installer from python.org/downloads/macos and run the .pkg file.

Step 2 — Place the Optima folder

Unzip the Optima.zip file you received into a stable location. I recommend:

+ **macOS:** ~/Applications/Optima

The folder should look like this once unzipped:

```
Optima/  
├── app/ # the application code  
├── prototypes/ # earlier prototype snapshots (read only)  
├── docs/ # this documentation suite  
├── feedback/ # evaluator responses  
├── run.py # the launcher script  
├── requirements.txt # pinned dependency list  
├── img/  
└── README.me
```

Step 3 — Installing Python 3.11+

A virtual environment keeps Optima's libraries separate from anything else on your system. This step is optional, but still recommended, especially on macOS, where modern Python refuses to install packages.

From inside the Optima/ folder:

```
# macOS  
cd ~/Applications/Optima  
python3 -m venv .venv  
source .venv/bin/activate
```

You'll know the virtual environment is active when your terminal prompt starts with (.venv).

Step 4 — Installing dependencies

With the virtual environment active:

```
pip install -r requirements.txt
```

This installs two libraries.

Library	Why
Flask ≥ 3.0	Routes the URLs inside the desktop window.
pywebview ≥ 5.0	Wraps Flask in a native window so it runs as a desktop app.

Expect this step to take 30-90 seconds, depending on your connectivity.

Step 5 — Run the application

Still inside the Optima/ folder, with the virtual environment active:

```
python3 run.py
```

If that doesn't work, try:

```
python run.py
```

A native window titled **Optima — Automated Study Flow** should open within a few seconds, displaying the splash screen.

Optional flags

Flag	Effect
--browser	Skip the native window and open Optima in your default web browser at http://127.0.0.1:5050/ instead. Useful if PyWebview is misbehaving on your platform.
--port 5060	Run on a different port if 5050 is already taken.
--host 0.0.0.0	Bind to all interfaces. This is not normally needed, only if you want to test from another device on your local network.

Expect this step to take 30-90 seconds, depending on your connectivity.

MAKING A DESKTOP SHORTCUT

To launch Optima with a single click from your dock rather than a terminal command, see the *User Manual* > §13. *Adding a Desktop Shortcut* section. In short, it uses Apple's native Shortcuts application to execute `python3 run.py` from an application icon.

Step 6 — First-time account setup

1. On the splash screen, click **Create a new account**.
 2. Enter a username (3-32 characters, letters and numbers, plus . - _ @) and a password (6+ characters). Confirm the passwords.
 3. Click **sign up**.
 4. You'll land on the dashboard with a starter set of subjects already seeded. Replace and edit them from the preferences page.
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Troubleshooting

"command not found: python3"

On macOS, try 'python' instead, or reinstall using Homebrew or via the official website.

"externally-managed-environment" error during pip install

You skipped step 3. Modern macOS may block global installs, so go back to step 3 and create a virtual environment, then try again.

The screen opens black or grey

Pyview's renderer didn't initialise. Restart, or try running with `python3 run.py --browser` and Optima will open in your normal browser instead.

"Port 5050 is already in use."

Another application is already using port 5050. Run with `python3 run.py --port 5060` (or any free port).

I forgot my password.

Skill issue. Optima has no password reset, as there's no email field during the sign-up process. You have two options:

- + **Create a new account** and start fresh.
- + **Copy the user .json** data into a new account (see User Manual FAQ for further details).

The application froze

If the window is unresponsive for more than 10 seconds, force quit it (Cmd + Opt + Esc). Your data will always remain completely written and automatically saved. Relaunch, and the latest saved version will open.

Uninstalling/Wiping data

To uninstall completely, delete the entire Optima/ folder. There are no system-wide files to erase.

To remove just one account while keeping the application installed, delete that account's JSON file from data/users.